

Pranay Karakoti

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EDUCATION

University of Illinois at Chicago

Chicago, IL

B.S., Computer Science

May 2027

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Discrete Math, Operating Systems, Machine Organization, Software Design, Database Systems, Artificial Intelligence I, Computer Design

EXPERIENCE

FlyRank AI

Jul 2026 – Sep 2026

Back-End AI Engineering Intern

Chicago, IL

- Enabled the team to switch between 3 AI providers (Groq, Gemini, Ollama) with a single config change and 0 application code changes, by designing a provider-agnostic integration layer in Python with a shared interface and Pydantic-typed responses.
- Kept the AI summarization service at 0 failures across a 10-case reliability suite, including malformed and empty model responses, by adding automatic schema validation, single-retry fallback, and per-request cost telemetry in FastAPI.
- Eliminated data loss on restarts and outages, verified across 100% of records by automated recovery tests, by migrating storage from in-memory to PostgreSQL with SQLAlchemy and containerizing the service in Docker for identical behavior across environments.

100 Devs

Jun 2025 – Jun 2026

Software Engineer

Remote

- Automated tracking for 50+ job applications, cutting manual tracking time by 50% and replacing error-prone spreadsheets, by building a full-stack web app with FastAPI, PostgreSQL, and SQLAlchemy.
- Reduced API integration errors by 50-60% across 7 endpoints as measured during testing by adding schema validation and structured logging using FastAPI and Pydantic.
- Delivered fully automated, zero-touch deployments to AWS, eliminating all manual release steps, by building a GitHub Actions pipeline with Docker image builds to ECR, OIDC authentication, and AWS Secrets Manager for credentials.

PROJECTS

M.A.C.E. (Monolith Analysis & Clustering Engine) | *Python, LangChain, Neo4j, FastAPI, Docker*

[GitHub](#)

- Cut codebase analysis time by 30-40% as measured by manual review benchmarks by building a Python tool that analyzes dependencies and suggests service boundaries using LangChain and Neo4j.
- Engineered a Python-based system to duplicate and compare live application requests across 6+ key features, catching inconsistencies and reducing deployment-related errors by 20%.
- Generated 3-5K lines of FastAPI service code, reducing initial setup time from several hours to under 30 minutes by developing an AI-assisted code generation pipeline using Python, LangChain, Docker, and GitHub Actions.

Fraud & Anomaly Detector | *Python, XGBoost, PyTorch, scikit-learn*

[GitHub](#)

- Achieved a PR-AUC of 0.876 on validation (0.836 on test, 82.43% recall) detecting fraud in a highly imbalanced 284K-transaction dataset (0.172% fraud rate) by tuning an XGBoost classifier with class weighting.
- Compared six modeling approaches, including weighted logistic regression, three XGBoost variants (weighting, SMOTE, undersampling), and a PyTorch autoencoder, using PR-AUC instead of accuracy to avoid misleading results on imbalanced data.
- Optimized the decision threshold against a \$5 per-transaction review cost via cost-sensitivity analysis, validated with stratified 70/15/15 splits and 5-fold cross-validation.

microvLLM | *LLM inference server | C++20, llama.cpp, CMake, CTest, Kubernetes, Prometheus*

[GitHub](#)

- Drove a 2.06x throughput improvement and 60% lower latency for short requests under mixed workloads by building a continuous-batching, chunked-prefill LLM inference server in C++20 over llama.cpp.
- Implemented a block-based KV-cache allocator with admission control and LIFO preemption, plus prefix sharing via copy-on-write, to keep the system correct under tight cache budgets during concurrent requests.
- Built and deployed a C++ LLM inference service on Kubernetes, configuring health checks, Prometheus monitoring, resource limits, and scalable replicas for concurrent inference workloads.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript, HTML/CSS

Frameworks & Tools: Kubernetes, Docker, Git, GitHub Actions, FastAPI, React.js, Node.js, Streamlit, OpenCV, PostgreSQL, Pandas, MongoDB

AWS / Cloud: App Runner, ECR, RDS, S3, CloudFront, IAM (OIDC), Secrets Manager, VPC

AI / ML: LangChain, LangGraph, LLM APIs (OpenAI, Anthropic), RAG, Prompt Engineering, XGBoost, Vector DBs, Pydantic, ControlNet, PyTorch